AZURE ENVIRONMENT SETUP, IDENTITY MANAGEMENT & RBAC IMPLEMENTATION (HOMELAB)

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1. PROJECT OVERVIEW

This project demonstrates Azure cloud administration tasks within a controlled Microsoft Azure environment, focusing on infrastructure deployment, identity management, automation, and access control.

The implementation includes the provisioning of cloud resources using Azure CLI, deployment of a Windows Server virtual machine, and configuration of identity services through Microsoft Entra ID. User accounts and security groups were created to simulate a structured organizational environment.

To improve efficiency and scalability, dynamic group membership was configured based on user attributes, allowing automatic assignment of users to appropriate groups without manual intervention. In addition, Role-Based Access Control (RBAC) was implemented using custom roles to manage administrative permissions in a controlled manner.

The project reflects practical cloud administration workflows commonly used in enterprise environments, particularly in managing resources, identities, and access policies within Microsoft Azure.

2. OBJECTIVES

- Deploy and configure core Azure infrastructure using Azure CLI
- Implement a structured cloud environment using resource groups and virtual machines
- Configure identity management using Microsoft Entra ID
- Apply role-based access control (RBAC) for permission management
- Automate user access through dynamic group membership rules
- Validate identity and access configurations through testing

3. ENVIRONMENT & TOOLS

- Microsoft Azure – Cloud platform for deployment and management
- Microsoft Entra ID – Identity and access management
- Azure CLI – Command-line automation tool
- Windows Server 2016 (MoonTechVM) – Virtual machine operating system
- Resource group management
- Virtual machine deployment
- User and group management
- Dynamic group membership
- Role-Based Access Control (RBAC)

4. IMPLEMENTATION DETAILS

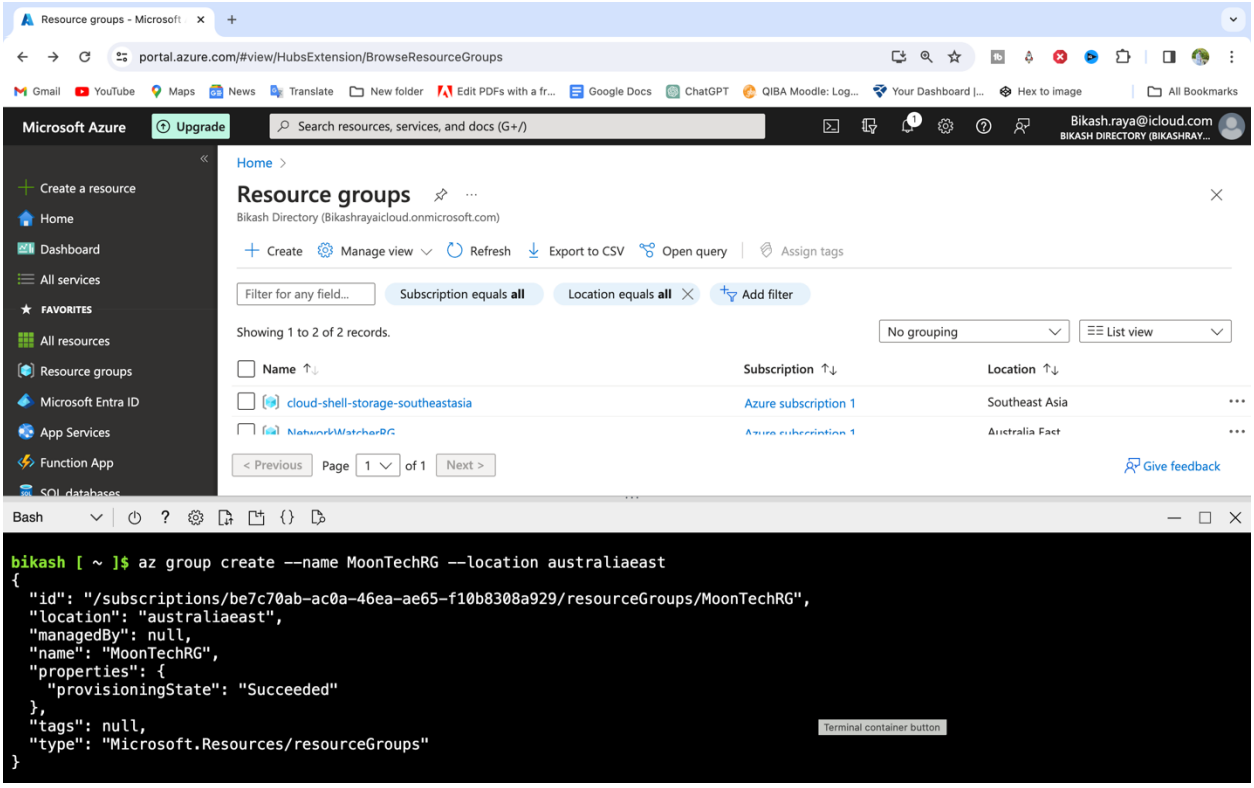
4.1 RESOURCE GROUP CREATION

The first step involved creating a resource group in Microsoft Azure named **MoonTechRG**.

Using Azure CLI, an Azure resource group named **"MoonTechRG"** was created in the **Australia East** region.

This resource group acted as the primary logical container for all resources deployed within the MoonTech Azure environment, enabling centralized management, monitoring, and governance.

Screenshots: Resource Group Creation

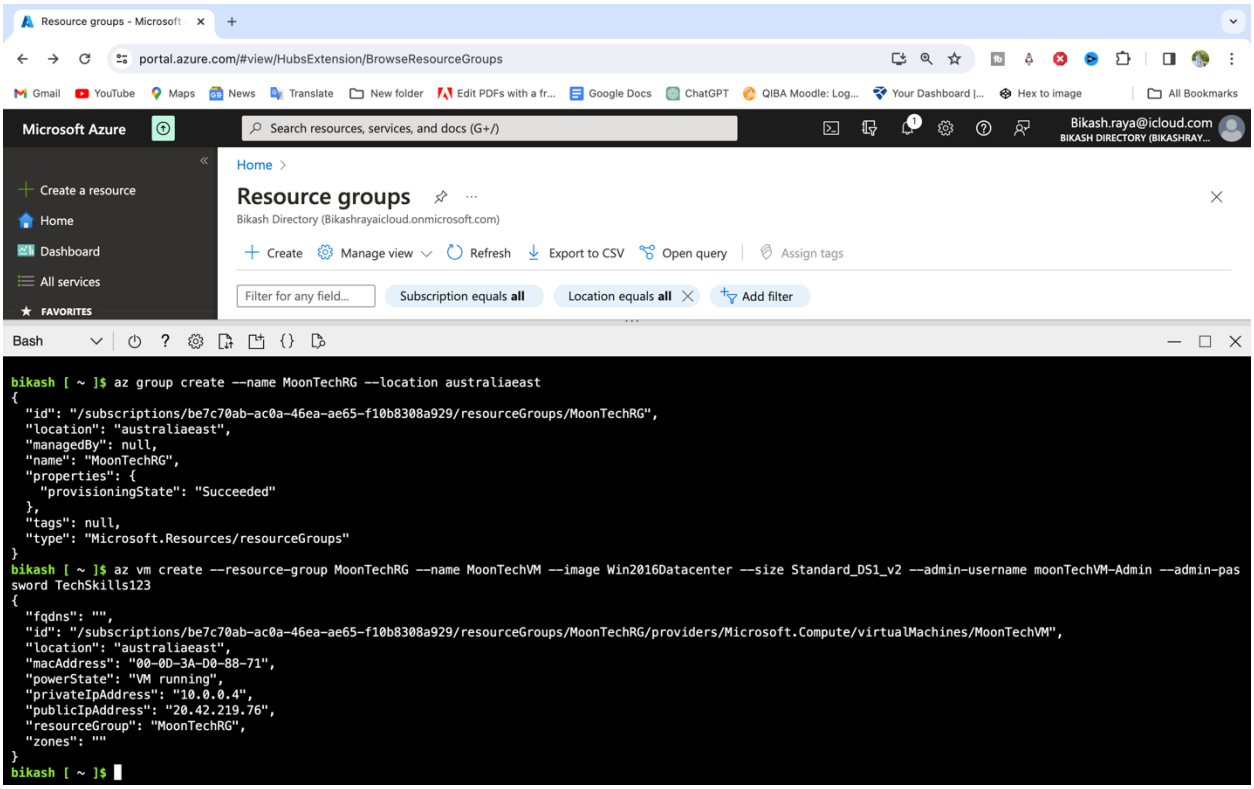


4.2 VIRTUAL MACHINE DEPLOYMENT (MOONTECHVM):

• Using Azure CLI, a Windows Server 2016 Standard virtual machine named **"MoonTechVM"** was deployed within the **"MoonTechRG"** resource group.
• An appropriate virtual machine size was selected based on expected workload and performance requirements.
• Networking components including virtual network, subnet configuration, and security settings were configured during deployment.
• Storage was provisioned using managed disk configuration to ensure reliability and scalability.

The virtual machine served as the primary compute resource for system administration and simulated enterprise server operations within the Azure environment.

📸 Screenshots: Virtual Machine Deployment



4.3 USER AND GROUP MANAGEMENT

User accounts for **Tim, Jack, Ronaldo, and Messi** were created in Microsoft Entra ID using Azure CLI.

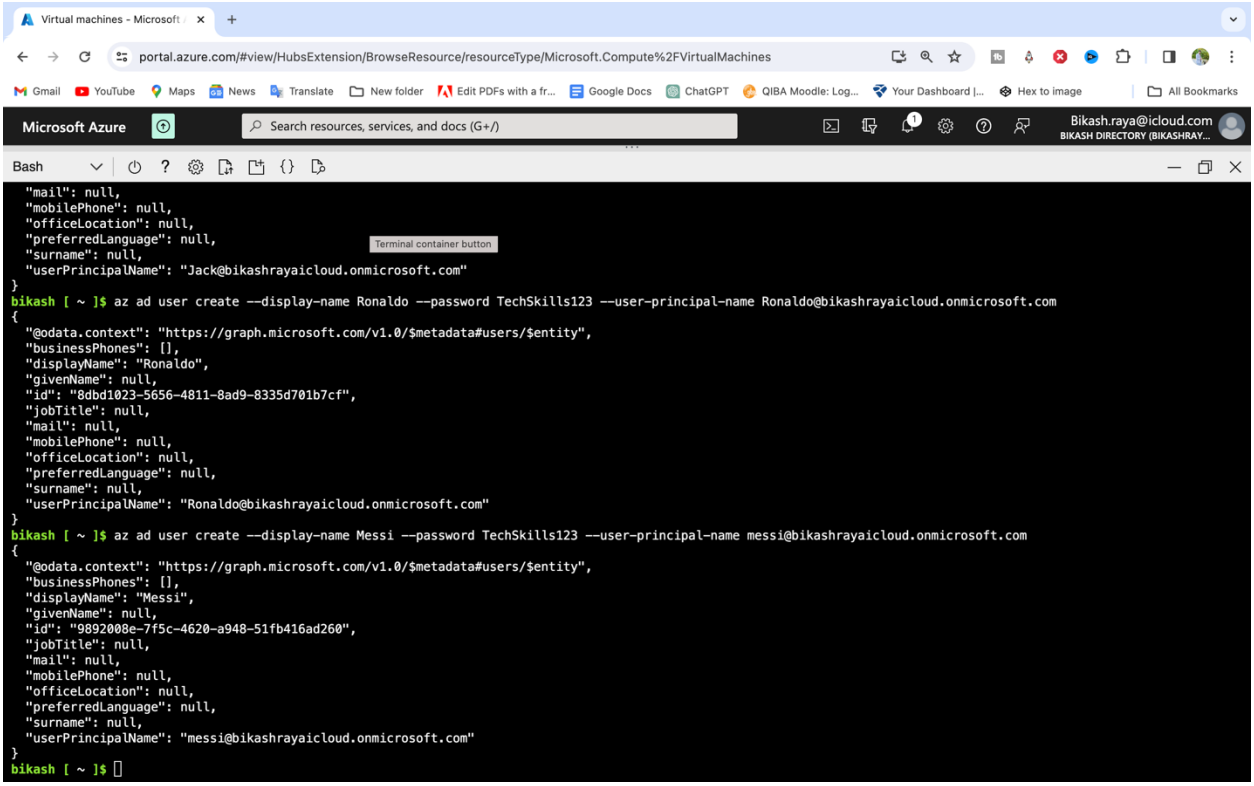
Two security groups were then created:

- **MoonTechSales**
- **MoonTechMarketing**

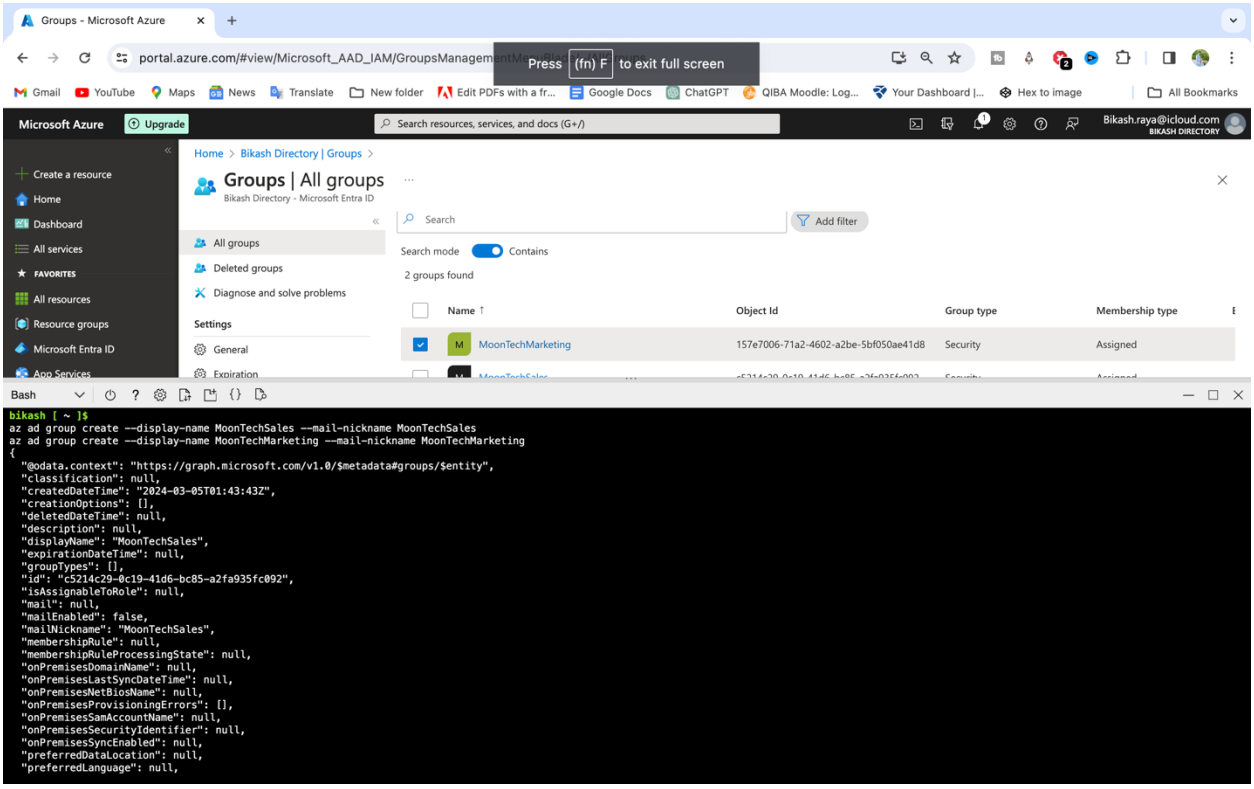
These groups were used to represent different departments and to organize users accordingly.

📸 Screenshots: Manage User & Groups

Creating Users:



Creating Groups:



4.4 DYNAMIC GROUP CONFIGURATION

Dynamic group membership rules were implemented for **"MoonTechSales"** and **"MoonTechMarketing"** based on user attributes within Microsoft Entra ID.

The Department attribute was used as the condition for automated group assignment.

User attributes were configured as follows:

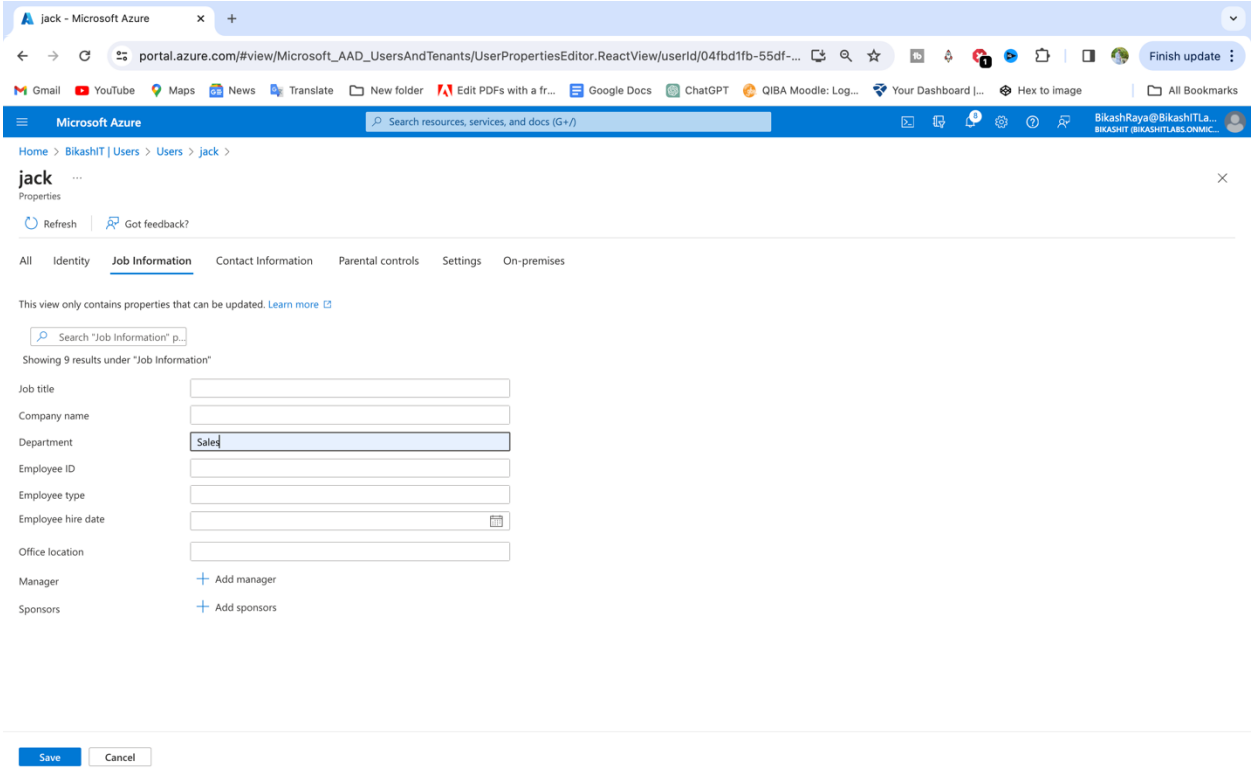
- Tim → Sales
- Jack → Sales
- Ronaldo → Marketing
- Messi → Marketing

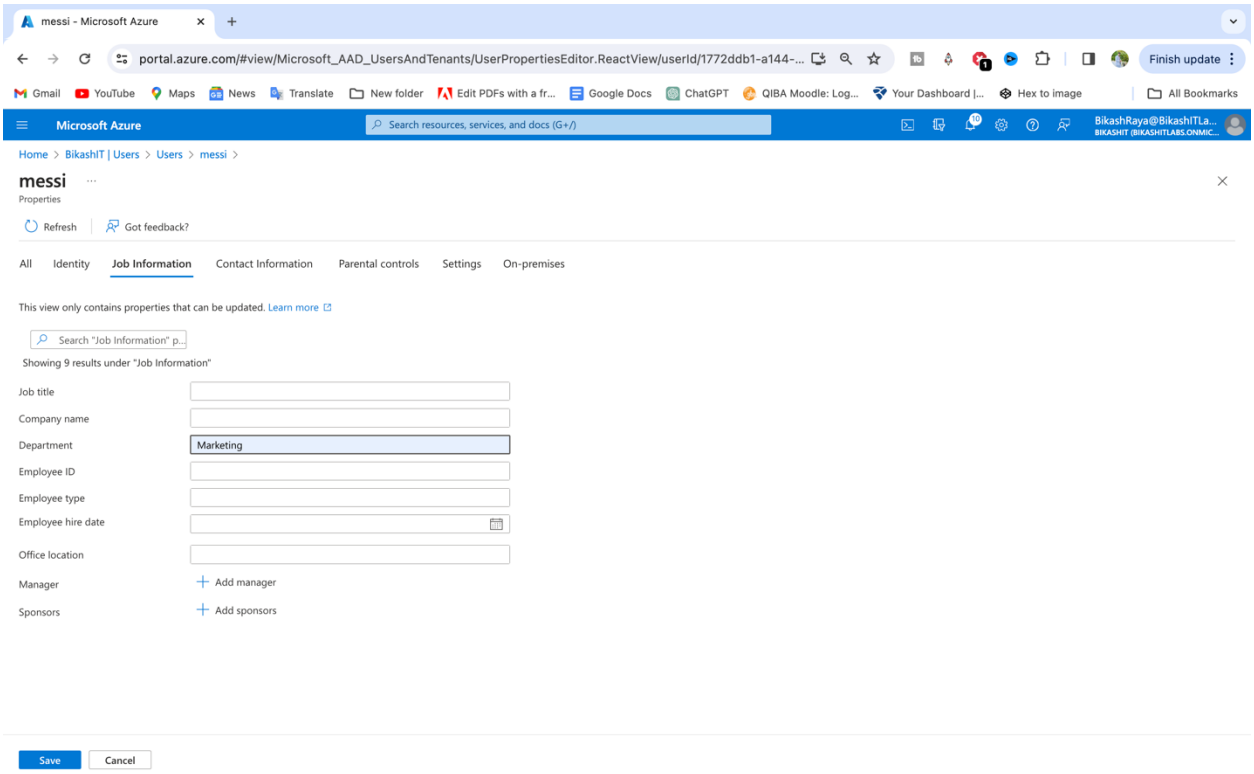
Based on these attributes, users were automatically assigned to their respective groups:

- Tim and Jack were automatically added to **MoonTechSales**
- Ronaldo and Messi were automatically added to **MoonTechMarketing**

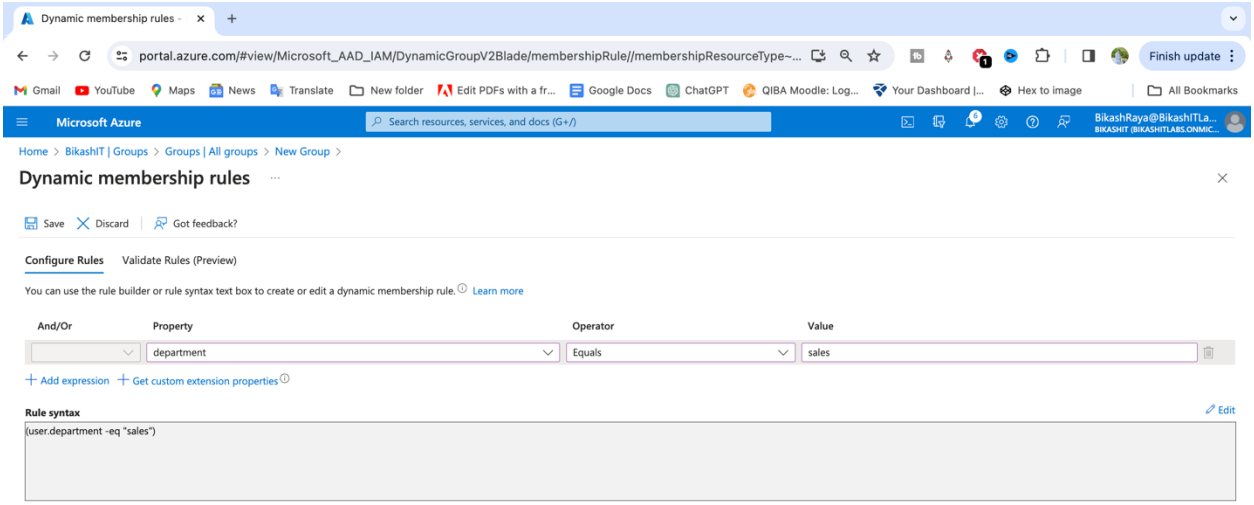
This implementation ensured automated identity governance and eliminated manual group assignment overhead.

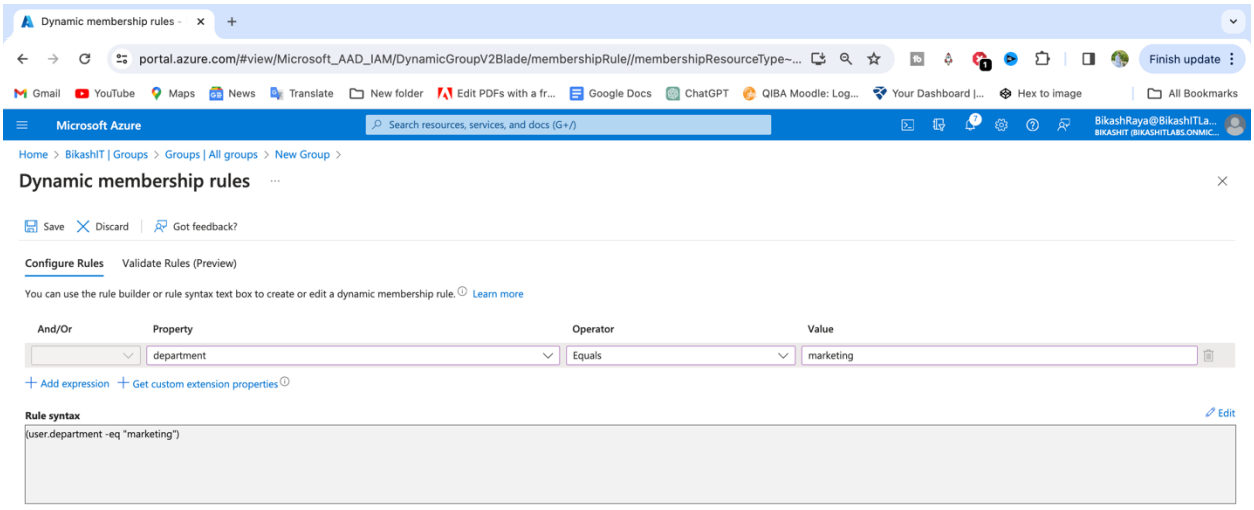
📸 *Screenshots: Dynamic Group Membership Rules Configuration*





Setting dynamic group membership rules for MoonTechSales and MoonTechMarketing based on user attributes “sales” and “marketing”:

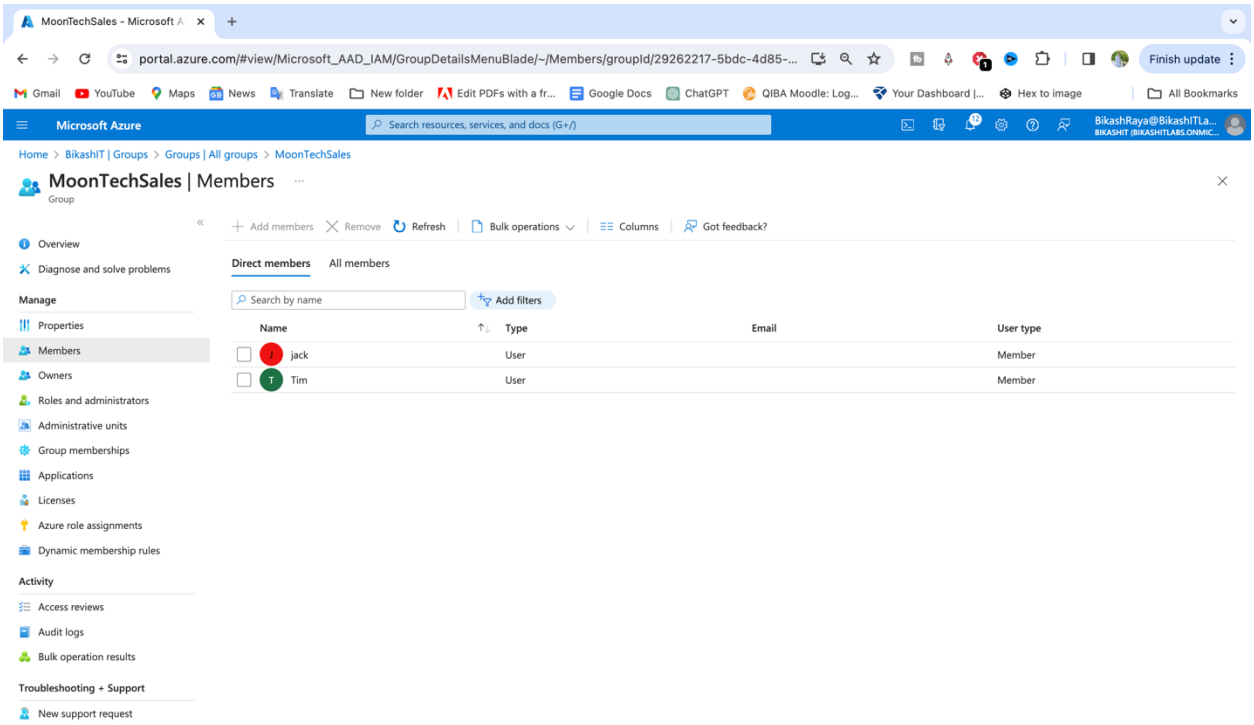


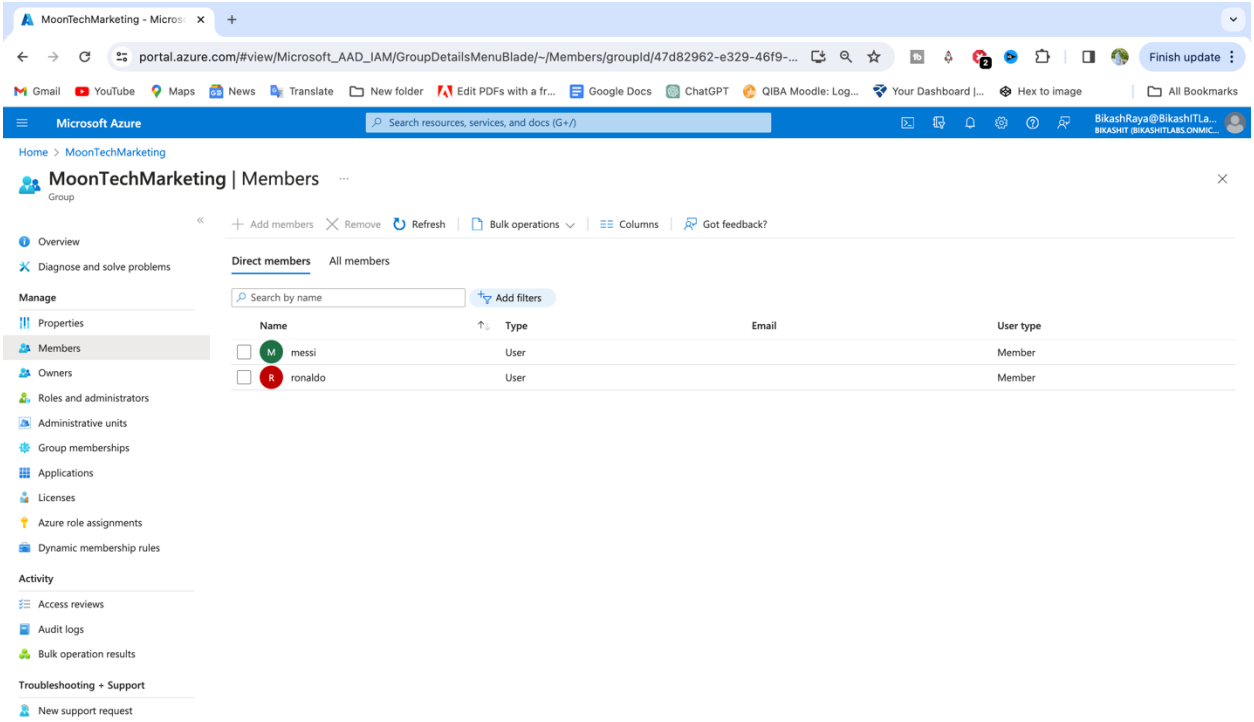


4.5 VALIDATION OF DYNAMIC ASSIGNMENT

Group membership was validated to confirm that Tim and Jack were automatically added to the MoonTechSales group, while Ronaldo and Messi were assigned to the MoonTechMarketing group based on the defined dynamic rules

Screenshots: Validation of Dynamic Group Membership Assignment





4.6 ROLE-BASED ACCESS CONTROL (RBAC) CONFIGURATION

A custom RBAC role named **SalesAdmin** was created with permissions to manage user assignments within the MoonTechSales group. This role was assigned to Ronaldo.

Another custom role named **MarketingAdmin** was created with permissions for managing the MoonTechMarketing group. This role was assigned to Messi.

These configurations demonstrate controlled delegation of administrative access.

Screenshots: RBAC Role Creation and Role Assignment (SalesAdmin & MarketingAdmin)

New custom role - Microsoft

portal.azure.com/#view/Microsoft_AAD_IAM/CreateCustomRoleBlade/isCreateMode~/true

Microsoft Azure

Home > BikashIT | Roles and administrators > Roles and administrators | All roles >

New custom role

All roles

Got feedback?

Basics | Permissions | Review + create

Below is the definition of your role. Click Create to save it and then you can assign members to it.

Name: SalesAdmin

Description:

Permission	Description	Privileged
microsoft.directory/groups.security.assignedMembership/members/update	Update members of Security groups of assigned membership type, excluding role-assignable groups	

Previous | Create

Add assignments - Microsoft

portal.azure.com/#view/Microsoft_AAD_IAM/RoleMenuBlade/~/Role...

Microsoft Azure

Home > BikashIT | Roles and administrators > Roles and administrators |

SalesAdmin | Assignments

All roles

Diagnose and solve problems

Manage

Assignments

Description

Activity

Bulk operation results

Troubleshooting + Support

New support request

Add assignments

Search: Search by name

Type: All

Name

No role assignments found

Add assignments

Try changing or adding filters if you don't see what you're looking for.

Only groups eligible for role assignment are displayed.

Learn more

Search

5 results found

All | Users | Groups

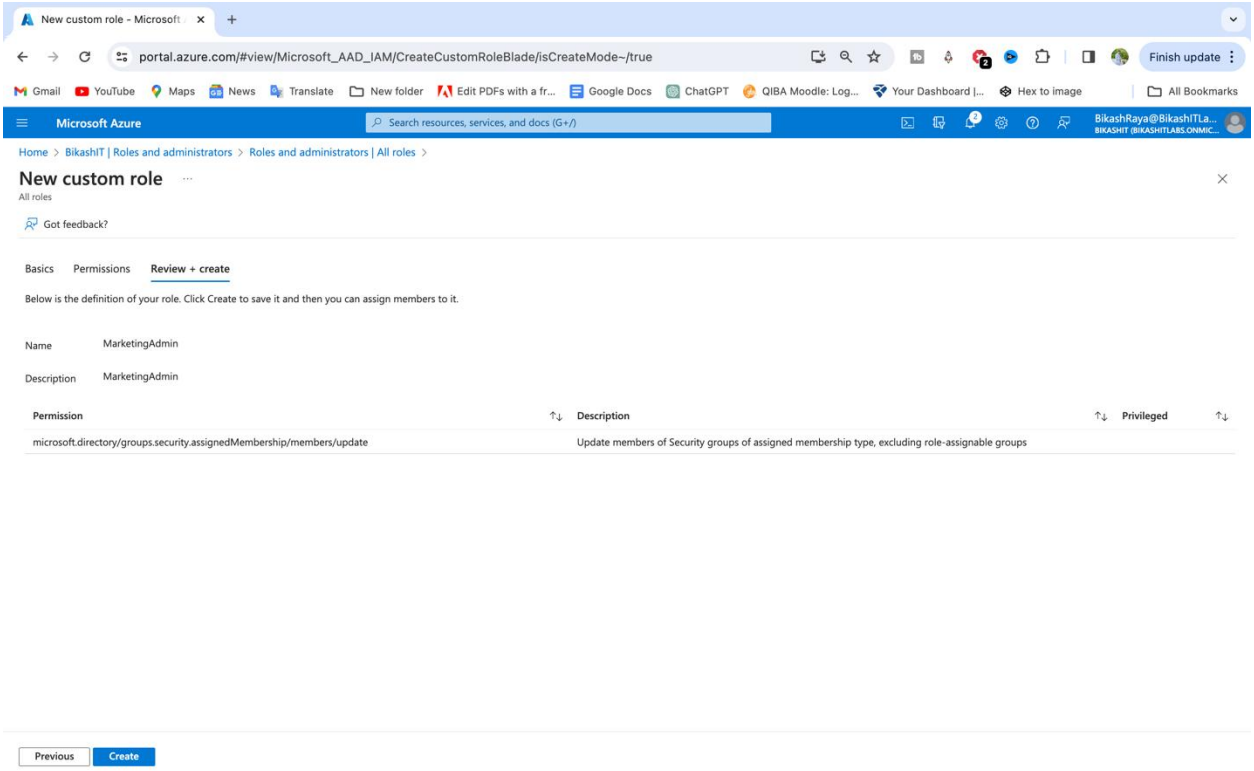
Name	Type	Details
Bikash Raya	User	BikashRaya@BikashITLabs.onmicrosoft.com
jack	User	jack@BikashITLabs.onmicrosoft.com
messi	User	messi@BikashITLabs.onmicrosoft.com
ronaldo	User	ronaldo@BikashITLabs.onmicrosoft.com
Tim	User	Tim@BikashITLabs.onmicrosoft.com

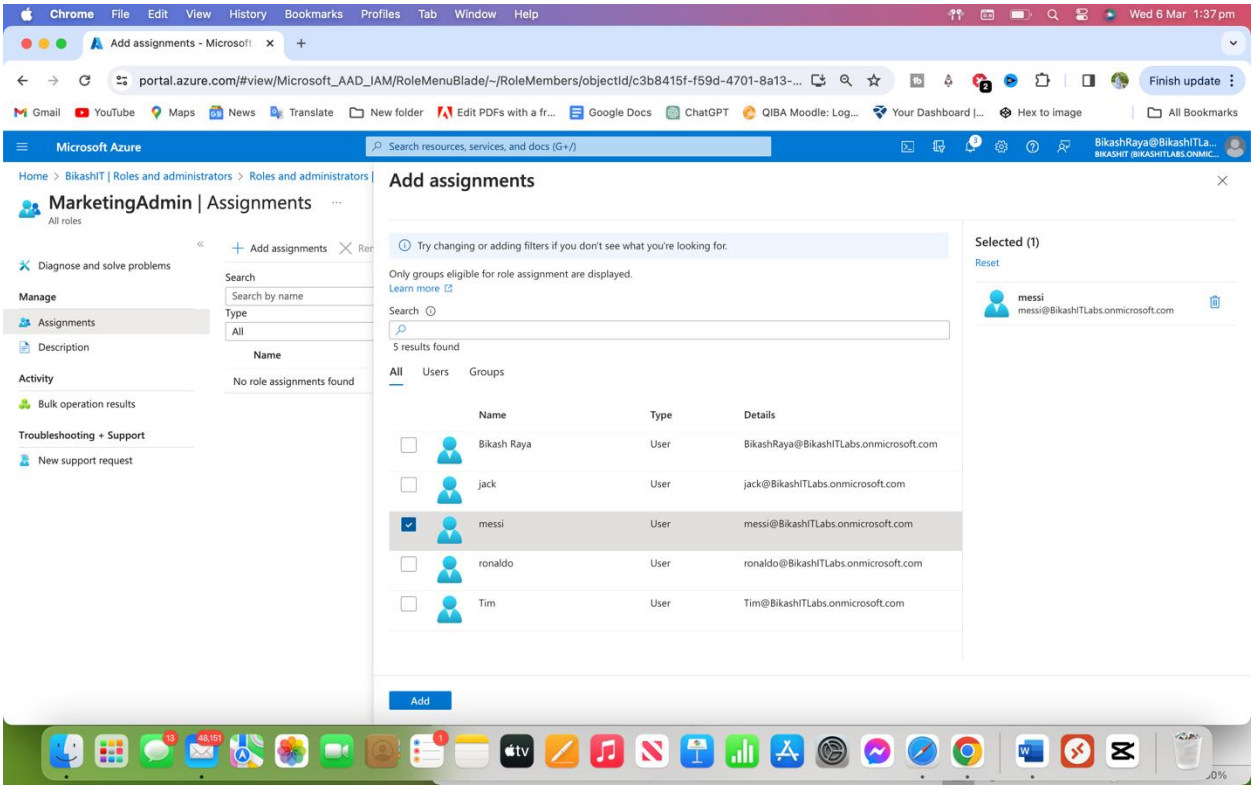
Selected (1)

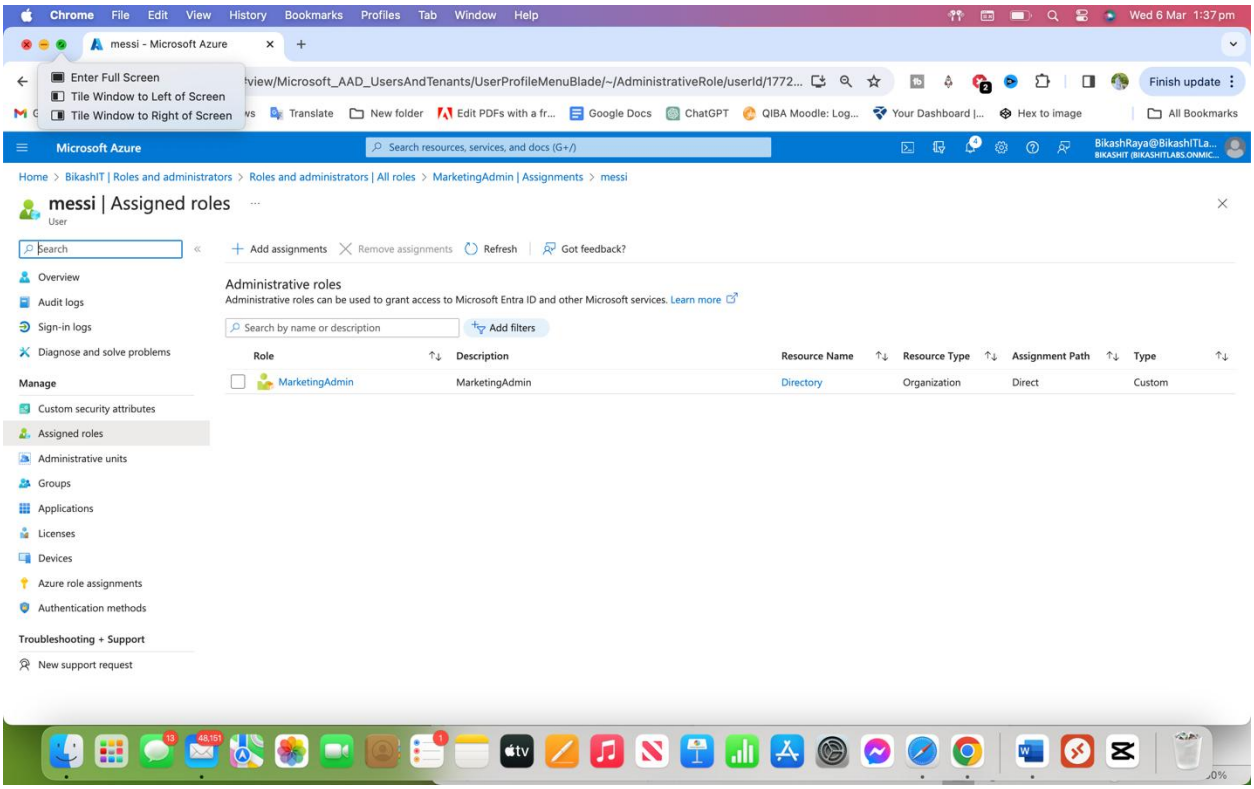
Reset

ronaldo
ronaldo@BikashITLabs.onmicrosoft.com

Add


New custom role - Microsoft
portal.azure.com/#view/Microsoft_AAD_IAM/CreateCustomRoleBlade/isCreateMode~/true
Finish update
Microsoft Azure
Search resources, services, and docs (G+/)
Home > BikashIT | Roles and administrators > Roles and administrators | All roles >
New custom role
All roles
Got feedback?
Basics
Permissions
Review + create
Below is the definition of your role. Click Create to save it and then you can assign members to it.
Name
MarketingAdmin
Description
MarketingAdmin
Permission
Description
Privileged
microsoft.directory/groups.security.assignedMembership/members/update
Update members of Security groups of assigned membership type, excluding role-assignable groups
Previous
Create


Add assignments - Microsoft
portal.azure.com/#view/Microsoft_AAD_IAM/RoleMenuBlade/~/RoleMembers/objectId/c3b8415f-f59d-4701-8a13-...
Microsoft Azure
Home > BikashIT | Roles and administrators > Roles and administrators |
MarketingAdmin | Assignments
All roles
Diagnose and solve problems
Manage
Assignments
Description
Activity
Bulk operation results
Troubleshooting + Support
New support request
Add assignments
Search
Search by name
Type
All
Name
No role assignments found
Add assignments
Try changing or adding filters if you don't see what you're looking for.
Only groups eligible for role assignment are displayed.
Learn more
Search
5 results found
All
Users
Groups
Name
Type
Details
Bikash Raya
User
BikashRaya@BikashITLabs.onmicrosoft.com
jack
User
jack@BikashITLabs.onmicrosoft.com
messi
User
messi@BikashITLabs.onmicrosoft.com
ronaldo
User
ronaldo@BikashITLabs.onmicrosoft.com
Tim
User
Tim@BikashITLabs.onmicrosoft.com
Selected (1)
Reset
messi
messi@BikashITLabs.onmicrosoft.com
Add



5. CONCLUSION

This project demonstrates the successful setup of a Microsoft Azure environment, including resource provisioning, virtual machine deployment, identity management, dynamic group automation, and RBAC implementation.

The implementation reflects key cloud administration practices, including structured resource management, identity governance, and secure access control in Azure environments.